

Baby Hoover V2

Operator Guide — live capture and offline replay

Instrument: T11_F125_Baby_Hoover_V2_15SEP26.py · For: Mike (RSO / capture PC) · 15 SEP 26

What it is, in one paragraph

Baby Hoover V2 does two jobs from one code path. **Live**, on the capture PC, it records every datagram to a .bin, directs the in-game spectator camera, and writes a timed two-voice script. **On replay**, it reads a captured .bin back deterministically (no game, no socket, no wall clock) and produces the identical set of artifacts. Nothing synthesises audio or calls a model: V2 writes a script, a human pastes it into ElevenLabs afterwards. Every run leaves four artifacts plus a pronunciation lexicon. Every scoring number lives in hoover_config_v2.json, which is hashed and stamped into every artifact.

1 Requirements

- **Python 3.8+**, standard library only. No pip install, no dependencies.
- **Live camera control needs Windows** (SendInput scancode). On any OS you can still record and run advisory mode; on non-Windows the camera auto-disables.
- Files that travel together, in the repo root: **T11_F125_Baby_Hoover_V2_15SEP26.py** (the instrument), **hoover_config_v2.json** (every tunable — keep it beside the script), **hoover_roster_league.json** (the league identity roster).

2 Live run (capture PC, league night)

Telemetry settings on every driver's console (mirror the race card): UDP Telemetry ON · UDP Send Rate 60 Hz · UDP Format **2025** · Your Telemetry **Public** · Show Online Names **ON**. The last two are what let the booth name people and read tyres/fuel; a driver who joins with names off is stuck at “the car” for the whole night.

Then on the capture PC:

```
python3 T11_F125_Baby_Hoover_V2_15SEP26.py --roster hoover_roster_league.json
```

The rig prints a banner (config hash, camera state), then:

- **Start OBS recording.** Press ENTER when it is rolling.
- You get ~10 s to **click the F1 25 window**. After that **do not alt-tab** — SendInput goes to the foreground window and the game must hold focus. If the camera gets moved by hand, the rig detects it and yields for 20 s rather than fighting.
- Leave it running. Sessions roll automatically; each is finalised into its own folder named from what the game actually said (e.g. 03_Austria_Race).
- Ctrl-C at the end, or let it idle-close after 90 s without lap data.

Useful flags: --no-camera (advisory only), --walk-fallback (re-enable the F7 relative walk after a failed direct select; off by default), --port N (default 20777), --outdir DIR, --run-id NAME, --note "text".

Focus is the whole ballgame. Every missed cut on 8 SEP traced to focus loss. One screen, one game, no clicking the console.

3 Replay a captured bin (tuning, or reproducing s11 / s05)

No game, no OBS, no camera. Deterministic — same bin + same config = byte-identical artifacts.

```
python3 T11_F125_Baby_Hoover_V2_15SEP26.py \  
  --replay HOOVER_20260908_202102_s11.bin \  
  --roster hoover_roster_league.json \  
  --outdir ./out --run-id s11
```

Output lands in ./out/s11/<NN_Track_Type>/. Point --replay at any .bin written by V1, V2, or T8V1_Recorder — the container framing is the same. To change behaviour, edit hoover_config_v2.json and re-run: the config hash changes, every artifact records the new hash, and the decision log shows the change. That is how you tune by ear and know exactly what moved.

4 Checks after a replay

Tier A detectors (must all read zero). Point at the artifact folder and stem:

```
python3 T11_F125_Baby_Hoover_V2_15SEP26.py \  
  --detectors "./out/s11/01_Austria_Race:s11_s01"
```

Prints each detector's hits and a TIER A TOTAL. Zero is a pass; the exit code is non-zero if anything fired. **A detector is never relaxed to make output pass** — if one fires and the output genuinely sounds fine, the declaration changes in writing, with a reason, in the detector's docstring.

Before / after table (V1 s11 as the baseline vs a V2 replay):

```
python3 tests/measure_before_after.py before <v1_s11_dir> HOOVER_20260908_202102_s11  
python3 tests/measure_before_after.py after ./out/s11/01_Austria_Race s11_s01
```

Reports peak demand, dead air, opener novelty, human camera share, utterance median, short-tail share, and connective-tissue rate.

Unit tests (9 checks: naming corpus, detectors, determinism):

```
python3 tests/test_baby_hoover_v2.py
```

Regression: when a replay sounds right, keep that config + output as the known-good snapshot; every later change diffs against it. Replay is deterministic, so this is nearly free.

The measured V1 s11 baseline (the “before” column)

Metric	V1 s11 (before)	V2 target / direction
Peak channel demand	272%	bounded by the serial scheduler
Dead air	258 s	filled by the floor governor (<150 wpm is a fault)
Opener novelty (3-word)	0.40	higher (reported, not gated)
Human camera share	67.5%	rises with participation as a multiplier
Utterance median	8 words	~8 words, with a deliberate short tail

Metric	V1 s11 (before)	V2 target / direction
Short tail (<=5 words)	3.7%	~31%
Connective tissue	0.0 / min	~1.0 / min, only when true

5 Artifacts per session

All four are stamped with the config hash and anchored to lights out. “stem” is <run_id>_s<NN>.

File	What it is
<stem>_script.md	The draft two-voice script: air time from lights out, class, register, tense, word count, duration, truncation point.
<stem>_cuts.csv	Every camera cut: subject, hold vs floor applied, selection reason by term, the discard set with loss codes, participation multiplier, interrupt flag.
<stem>_beats.jsonl	The decision log, the primary instrument. Per line: winner scored term-by-term plus the top three losers with loss codes; the suppression log; silence accounting; a run summary (config hash, LGOT source, opener novelty).
<stem>_lexicon.json	Pronunciation dictionary for ElevenLabs (see section 7).

Also written: <stem>_preflight.json (roster resolution report), <stem>_manifest.json, <stem>_events.txt, and the recording <stem>.bin (live only).

6 The roster — editing it for the league

hoover_roster_league.json is the shared identity layer. Each driver has an assigned driver_id (permanent — everything season-long keys on it) and match fields (handle, race_number). The scaffold holds the six humans from s11 at **level B** (name derived from the gamertag by the ladder).

To upgrade a driver to level A — full first-and-surname intros, possessives, personal lines — set their spoken.full to the real name and level to “A”:

```
{ "driver_id": "d_ronin",
  "match": { "handle": "Ronin0700VII", "race_number": 77 },
  "level": "A",
  "spoken": { "short": "Alex", "full": "Alex Ronin",
    "possessive_ok": true },
  "participation": "human", "team": "Mercedes" }
```

- A handle change mid-season is a one-line edit (match.handle), not a broken record.
- The driver who ran as Player (car 19, no handle, race number 0) can only be matched by enabling online names next time, or by an operator assignment entry.
- AI cars are **not** in the roster — they resolve from telemetry to real surnames automatically.

7 ElevenLabs hand-off

- Load <stem>_lexicon.json as the pronunciation dictionary. It forces rung-4 letter codes to individual letters (IMN → “eye em en”), rung-5 numbers to words (77 → “seventy-seven”), and carries the authored IPA for the AI surnames the synthesiser mangles (Hülkenberg, Dürksen, Villagómez, Antonelli, Bortoleto). Entries are sorted longest-match-first; case-sensitive, first match wins.

- Synthesise each line in its assigned voice (LEAD = Northern play-by-play, ANALYST = Southern colour).
- Place each file on the CapCut timeline at its **air time offset from lights out**, aligned to the start lights in the OBS capture.
- Record actual vs estimated duration per line and send it back — that table refits the speech-rate estimator (currently 2.92 w/s).

8 One thing to watch: the lights-out anchor

The paper anchors every air time to the **LGOT** (lights-out) event. **Neither s11 nor s05 actually fired LGOT** — a known consequence of the capture being bound to a car that later left (the same reason race-end events go missing). V2 handles this: it uses LGOT when present, and otherwise **derives** lights-out from the first green race lap-data tick, stamping the source (lgot_source) into every artifact and the run summary.

When you align in CapCut, confirm the derived anchor lands on the start lights in the video. If it drifts by a constant amount, that offset is the correction — flag it and it becomes a config value.

9 What is NOT in V2 (by design)

No prediction loop, no audio interruption / ducking / stingers (em-dash notation only), no Tier 1 curated narratives (the slot interface exists and stays empty), no network / model / synthesis, no camera view-type assertion. Toddler Hoover picks those up; the writer already sits behind a clean `write_line(candidate, class, register, word_budget, facts)` seam, so swapping the slot grammar for a prompt assembler touches nothing else.

Repo files: T11_F125_Baby_Hoover_V2_15SEP26.py · hoover_config_v2.json · hoover_roster_league.json · tests/ (unit tests, replay-bin generator, before/after measurement). V1.1 preserved under archive/. Verified: all Tier A detectors zero, two same-config replays byte-identical, tests 9/9.